Inverted Transfer: A High-Frequency Account of Below-Chance Cross-Manipulation Deepfake Detection

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***Abstract*—Deepfake detectors trained on one manipulation method are known to generalize poorly to others, but some cross-manipulation cells on FaceForensics++ go further: they score *below* chance. An Xception detector trained on Deepfakes reaches a video-level AUC of 0.26 on FaceSwap. We show that this inversion is not a preprocessing artifact. A reference pipeline that changes the face detector, alignment, storage format, crop size and frame count reproduces it (0.26, 95% CI [0.21, 0.31], three seeds), and two architecturally different detectors fail in the same pattern ($r = 0.94$). We trace the inversion to high-frequency image energy: Deepfakes and NeuralTextures fakes are blurrier than the real faces they replace, FaceSwap fakes are sharper, and Face2Face fakes show no shift. The product of these per-method effect sizes predicts the observed transfer matrix (Spearman $\rho = 0.92$, best of all 24 method relabelings) and also predicts an independently published, official-data matrix ($\rho = 0.73$). Randomizing sharpness during training raises the worst cell by +0.098 AUC without an in-domain cost, but does not remove the inversion, indicating that high-frequency energy is a substantial but partial cause. The account is specific to FF++: on DeepFakeFace the same sign test fails.**

***Index Terms*—deepfake detection, generalization, shortcut learning, frequency analysis, FaceForensics++**

I. Introduction

Face-manipulation detectors routinely exceed 99% AUC when trained and tested on the same manipulation method, and lose much of that accuracy on methods they have not seen [1], [2]. The usual reading is that a detector learns method-specific artifacts and, on an unseen method, finds none and falls back to chance. Some cross-manipulation cells do not fit that reading: they land well *below* 0.5, meaning the detector ranks the unseen fakes as *more real* than real faces. A detector that is merely uninformed should not do that; one that has learned a cue whose sign flips between methods should.

We study this inversion on FaceForensics++ (FF++) [1], where it concentrates on FaceSwap (FS). Our contributions are:

- **Robustness of the effect.** The inversion survives a full change of preprocessing (detector, alignment, storage, resolution, frame sampling; 3 seeds, bootstrap CIs) and a one-factor-at-a-time preprocessing grid, and it is shared by structurally different architectures.
- **A measurable mechanism.** High-frequency energy shifts in opposite directions for FS and for Deepfakes (DF)/NeuralTextures (NT). The sign of this shift predicts which transfer cells invert, on our data and on independently published official-data results [2].
- **An intervention test.** Label-independent blur/sharpen randomization partially closes the inversion at no in-domain cost.

Below-chance scores on out-of-distribution data have been reported before and attributed to shortcut cues [3], [4]; our contribution is a within-FF++, manipulation-level account with a sign prediction that can be checked before any cross-manipulation test is run.

II. Related work

FF++ [1] introduced the four-method benchmark (DF, Face2Face F2F, FS, NT) and the Xception [5] baseline. Face X-ray [2] reported its cross-manipulation matrix: Xception trained on DF scores 0.49 on FS and NT-trained scores 0.48, i.e. at chance on official data, while its HRNet baseline falls below chance on the same cell. Frequency-domain artifacts of generated images are well documented [6], [7], and blending-based training such as Face X-ray and self-blended images [8] improves generalization by removing method-specific cues. Shortcut learning [3] gives the general frame: a classifier latches onto a cue that is predictive in training and need not be predictive, or even correctly signed, elsewhere.

III. Method

A. Data and protocol

We use FF++ c23 with the official video-level splits (720/140/140 identities for train/val/test) and label fakes as 1. **Data provenance:** all FF++ numbers come from a public Kaggle mirror of the dataset arranged in the official layout; official access was pending. We audited the mirror for real-versus-method confounds before training: container codec, profile, pixel format and x264 CRF are identical across all five classes, and face-detection failure is below 0.05% (Sec. V). The primary metric is video-level AUC (mean of frame probabilities), reported as the mean over seeds with a 95% CI from 1000 bootstrap resamples over test videos. A cell is called below chance only when its whole CI lies under 0.5.

B. Pipelines and models

The *original* pipeline uses MTCNN [9] without alignment, a $1.3\times$ margin, 20 frames per video and JPEG (q95) crops. The *reference* pipeline uses dlib HOG detection [10], 68-landmark

alignment [11] (eye line horizontal), a 1.3× margin, 256 px PNG crops and 32 frames per video. Detectors are ImageNet-pretrained Xception, ResNet-50 [12] and EfficientNet-B4 [13], all at 299 px input with 17.6–23.5 M parameters, plus a CNN-ViT hybrid (ResNet-50 stages 1–3 and a six-layer transformer [14], trained from scratch at 224 px). Training uses AdamW, cosine schedule, AMP, batch 32 and early stopping on validation AUC; each model is trained on a single manipulation plus real videos and tested on all four.

C. Image statistics and the sign test

For every test video we average, over eight frames, the high-frequency energy ratio h (share of Hann-windowed spectral energy above 0.25 cycles/pixel), Laplacian variance, a noise residual, JPEG blockiness, inner-face versus border colour and luminance differences and the gradient along the landmark-hull boundary. For method m we take the effect size d_m as Cohen's d of m's fakes versus real videos (positive: higher than real). If a detector trained on A relies on a statistic, its transfer to B should follow the sign of $d_A d_B$: same sign, the cue still points to "fake"; opposite sign, it points to "real" and the AUC inverts. We score $d_A d_B$ against the observed off-diagonal AUCs (Spearman ρ) and compare with all $4! = 24$ relabelings of the methods' d values, so the smallest attainable p is $1/24 \approx 0.042$. The test design was fixed before the statistics were examined.

IV. Results

A. The inversion and its robustness

Fig. 1 shows the transfer matrices. In the original pipeline, every cell pairing FS with DF or NT is at or below chance (DF→FS 0.26, NT→FS 0.32, FS→NT 0.40), while all pairings with F2F stay above chance. The reference pipeline, which changes every preprocessing choice at once, reproduces the matrix to within 0.05 AUC per cell: DF→FS is 0.26 [0.21, 0.31] and FS→NT 0.43 [0.37, 0.49], both below chance, while FS→DF (0.47 [0.41, 0.53]) is at chance.

Table I then varies single factors of the reference pipeline, each derived from the reference crops. Before any variant was evaluated, we measured each method's high-frequency effect size on each variant's crops and recorded predictions: JPEG q95, JPEG q75 and 10 frames leave the effect sizes almost unchanged, so the inversion should persist, while the tight 1.0× margin roughly halves them ($d_{DF} = -0.23$, $d_{FS} = +0.27$), so the inversion should weaken. DF→FS did not move under any variant (paired changes **TBD**, +0.032, +0.013, +0.001), so the tight-margin prediction failed for this cell: the cue the DF-trained detector uses survives removing the face border even though our summary statistic shrinks. The FS-trained row was the sensitive one. Under JPEG q75 and the tight margin, FS→NT rose (+0.044 [+0.013, +0.079]; +0.042 [+0.003, +0.084]) and FS→DF rose with it, the same pattern as the blur/sharpen intervention below. The JPEG q75 shift was not predicted. High-frequency energy therefore captures the *direction* of each transfer better than its *size*. The CNN-ViT hybrid, trained from scratch, fails in the same pattern as Xception across the 12 off-diagonal cells (Pearson $r = 0.94$, Spearman $\rho = 0.95$; Fig. 5), and Table III extends this to further backbones.

TABLE I
One-factor preprocessing grid (Xception, 2 seeds; reference pipeline otherwise). AUC, mean over seeds.

Variant	DF→FS	FS→NT	DF→DF
Reference (3 seeds)	0.26	0.43	1.00
JPEG q95	**TBD**	**TBD**	**TBD**
JPEG q75	0.29	0.47	1.00
10 frames/video	0.27	0.42	1.00
Tight 1.0× margin	0.26	0.47	1.00

B. Mechanism: high-frequency energy

High-frequency energy is the only statistic whose shift separates FS from DF and NT with every CI excluding zero, in both crop sets (Fig. 2): DF $d = -0.48$ [-0.72, -0.26], NT -0.39 [-0.61, -0.15], FS $+0.37$ [0.11, 0.66], F2F -0.01 [-0.23, 0.24]in the original crops, with the same pattern in the reference crops. Comparing each fake to the real video it was made from strengthens it (paired d: DF -0.58, NT -0.77, FS $+0.45$). DF and NT fakes are blurrier than real faces and FS fakes are sharper.

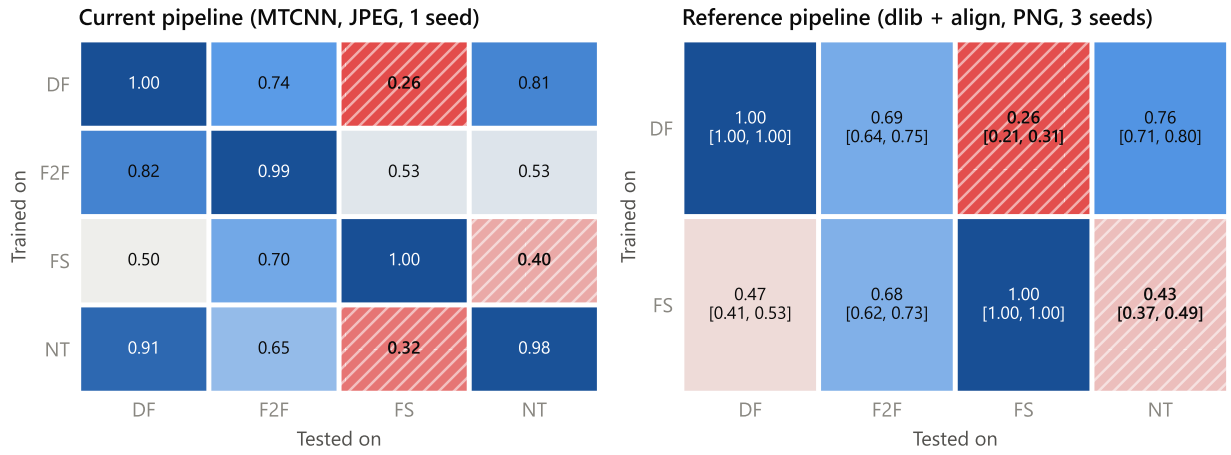
The sign test predicts the transfer matrices (Fig. 3): Spearman $\rho = 0.92$ for Xception and $\rho = 0.92$ for the hybrid, each the best of the 24 relabelings ($p = 0.042$), and $\rho = 0.90$ on the reference matrix. It mislabels only the F2F↔FS cells, where $d_{F2F} \approx 0$ carries no sign; the continuous score handles them. Correlated sharpness statistics (Laplacian variance, noise residual) predict nearly as well, but their FS shift is close to zero.

Two checks argue that detectors use this cue rather than merely co-varying with it. First, on real test faces alone, the DF-trained model's fake probability rises as faces get blurrier (Laplacian variance $\rho = -0.27$, $p = 0.001$; noise residual $\rho = -0.25$, $p = 0.002$), the direction DF's training fakes moved; the FS-trained model shows no significant reliance on high-frequency statistics. Second, the effect sizes measured on our mirror crops predict the Xception matrix published by Li et al. [2] on official data with $\rho = 0.73$ (best of 24 relabelings, $p = 0.042$).

C. Intervention

If high-frequency energy drives the inversion, removing it as a usable cue should reduce it. We train with label-independent random Gaussian blur or unsharp masking (probability 0.9), which moves the DF-versus-real effect size on training crops from -0.14 to 0.01 and FS from 0.42 to 0.27. Because the augmentation removes the DF cue but only weakens the FS one, we predicted beforehand that DF→FS would recover more than FS→NT. Table II gives the result, with paired bootstrap CIs on the change over the same test videos.

Every cell that pairs FS with DF or NT rises by about 0.1 AUC (DF→FS +0.098 [+0.070, +0.129], FS→NT +0.093 [+0.059, +0.129], FS→DF +0.105 [+0.071, +0.136])



Video-level AUC, Xception, FF++ c23 (Kaggle mirror). Hatched = below chance (reference: whole 95% CI < 0.5; current: point estimate < 0.47). Red < 0.5 < blue.

Fig. 1. Cross-manipulation video AUC for Xception. Left: original pipeline, one seed. Right: reference pipeline, three seeds (mean and 95% CI). Rows: training method; columns: test method. Hatched cells are below chance.

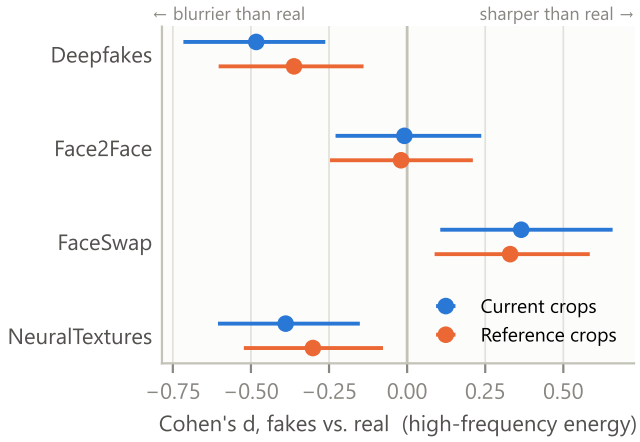


Fig. 2. High-frequency energy, fakes versus real (Cohen's d, 95% CI).

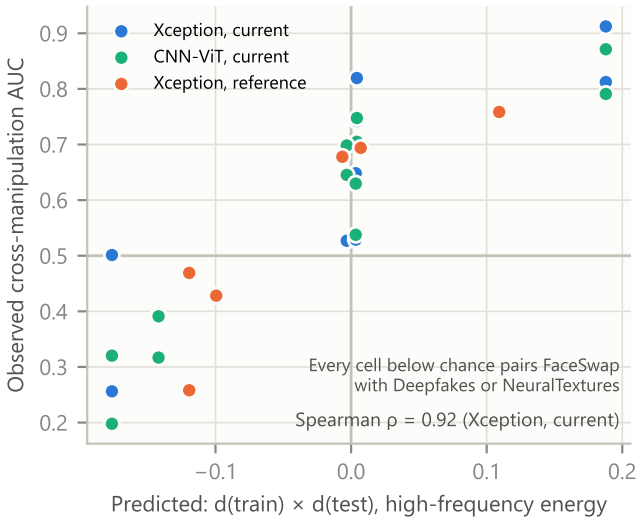


Fig. 3. Predicted $d_A d_B$ against observed cross-manipulation AUC.

with no in-domain cost. Both below-chance cells meet the success criterion we set in advance (CI excluding the reference mean): FS→NT is no longer below chance, but DF→FS remains inverted at 0.36 [0.30, 0.41]. Our directional prediction failed: the two cells recovered by the same amount. The augmentation is not free either: FS→F2F drops by -0.048 [-0.081, -0.015], suggesting that a FS detector denied the sharpness cue falls back on cues that transfer less well to F2F. We read this as high-frequency energy being a substantial, causal but partial contributor to the inversion.

D. Architectures

If the inversion comes from a property of the training fakes rather than of the network, other backbones should show it too. Table III repeats the reference-pipeline experiment with ImageNet-pretrained ResNet-50 and EfficientNet-B4 at the same 299 px input as Xception (EfficientNet-B4 at batch 16, the largest that fits in 6 GB). ResNet-50 reproduces Xception's below/at/above-chance call in 8/8 cells, including DF→FS at 0.29 [0.24, 0.35] and FS→NT at 0.42 [0.35, 0.48]; EfficientNet-B4 matches in 8/8.

E. Out-of-dataset check

On DeepFakeFace [15] (IMDB-WIKI reals; SD inpainting, InsightFace swaps and SD text-to-image fakes), we apply the same crops, measure d per generator, and evaluate the FF++-trained detectors without retraining (1000 images sampled per class; dlib found a face in 64–90% of them). All three generators are blurrier than DeepFakeFace's real images ($d = -0.26$, -0.52, -0.40), so the sign test predicts that DF-trained detectors transfer and FS-trained detectors invert on all three. It does not hold up (Table IV): the prediction is right on 2/4 cells

TABLE II
BLUR/SHARPEN RANDOMIZATION (3 SEEDS) VERSUS THE REFERENCE PIPELINE. Δ: PAIRED CHANGE WITH 95% CI.

Cell	Reference	Augmented	Δ
DF→FS	0.26	0.36	+0.098 [+0.070, +0.129]
FS→NT	0.43	0.52	+0.093 [+0.059, +0.129]
FS→DF	0.47	0.57	+0.105 [+0.071, +0.136]
DF→F2F	0.69	0.73	+0.040
FS→F2F	0.68	0.63	-0.048 [-0.081, -0.015]
DF→DF	1.00	1.00	0.000
FS→FS	1.00	1.00	0.000

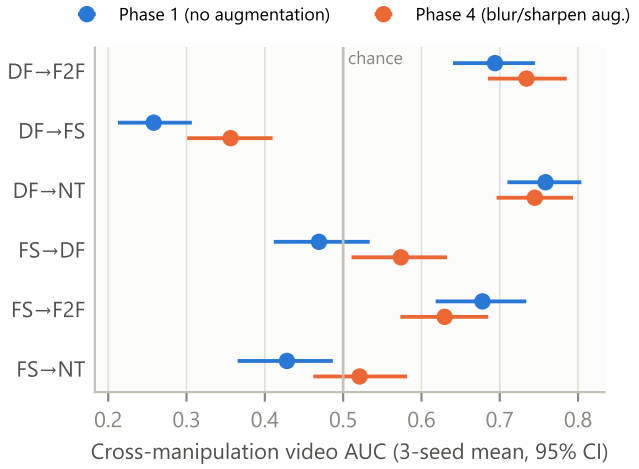


Fig. 4. Off-diagonal AUC with and without blur/sharpen randomization.

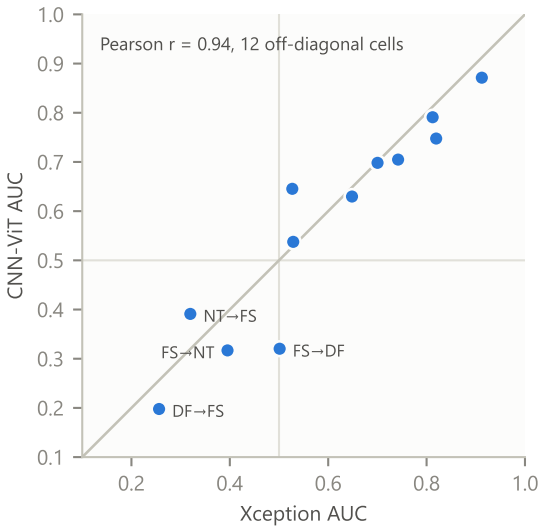


Fig. 5. Xception versus CNN-ViT transfer AUC (original pipeline).

TABLE III
BACKBONES ON THE REFERENCE PIPELINE (2 SEEDS; XCEPTION 3).

Backbone	DF→FS	FS→NT	DF→DF
Xception	0.26	0.43	1.00
ResNet-50	0.29	0.42	1.00
EfficientNet-B4	0.26	0.36	1.00

whose CI excludes chance, no better than a coin. SD text-to-image fakes score below chance for *every* detector, whatever the sign of its training cue. Across datasets, then, something other than high-frequency energy dominates; one candidate is that DeepFakeFace's real images (still photographs from IMDB-WIKI) are themselves out of distribution for detectors trained on YouTube video frames. The high-frequency account is a within-FF++ explanation.

V. LIMITATIONS

All FF++ results use an unofficial mirror; the agreement with published official-data results (Sec. IV-B) mitigates but does not remove this. The sign test has only four methods, so its permutation p cannot fall below $1/24$; we tested nine statistics, and the high-frequency statistics are correlated, so the mechanism is a well-supported candidate rather than a proof. Phase-1 and later models use 3 or 2 seeds; the original-pipeline matrices are single-seed point estimates. The architecture comparison in the original pipeline is confounded by pretraining and input size (the hybrid is trained from scratch at 224 px). Experiments ran on one 6 GB laptop GPU, which bounded the preprocessing grid to factors derivable from existing crops; detector and alignment were varied only jointly. On DeepFakeFace, face detection succeeded at different rates per class (64–90%), which may bias which images enter the evaluation.

VI. CONCLUSION

Below-chance cross-manipulation scores on FF++ are not noise or a preprocessing artifact: they follow the sign of a measurable high-frequency shift that differs between manipulation methods, a shift that predicts both our and published transfer matrices. The shift does not predict transfer to another dataset, where other differences dominate, so the account should be read as within-benchmark. Detectors should be checked for such signed shortcut cues before their cross-method scores are interpreted, and augmentation that randomizes the cue is a cheap partial fix.

DATA STATEMENT

FF++ is not redistributed; crops, checkpoints and per-video scores stay local. DeepFakeFace is used under its OpenRAIL licence with IMDB-WIKI-derived reals for research only. Code and all result tables are in the project repository.

REFERENCES

[1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[2] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, and B. Guo, "Face X-ray for more general face forgery detection," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

TABLE IV
FF++-TRAINED XCEPTION (REFERENCE PIPELINE, 3 SEEDS) ON DEEPFAKEFACE. AUC WITH 95% CI; REAL CLASS: IMDB-WIKI.

Generator	DF-trained	FS-trained
SD inpainting	0.46 [0.44, 0.49]	0.51 [0.49, 0.54]
InsightFace	0.65 [0.63, 0.68]	0.49 [0.46, 0.52]
SD text-to-image	0.32 [0.30, 0.34]	0.40 [0.38, 0.42]
Predicted	transfers	inverts

[3] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[4] Z. Yan *et al.*, "DF40: Toward next-generation deepfake detection," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.

[5] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[6] R. Durall, M. Keuper, and J. Keuper, "Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[7] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in *Proc. International Conference on Machine Learning (ICML)*, 2020.

[8] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[9] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.

[10] D. E. King, "Dlib-ml: A machine learning toolkit," *Journal of Machine Learning Research*, vol. 10, pp. 1755–1758, 2009.

[11] V. Kazemi and J. Sullivan, "One millisecond face alignment with an ensemble of regression trees," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[13] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. International Conference on Machine Learning (ICML)*, 2019.

[14] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. International Conference on Learning Representations (ICLR)*, 2021.

[15] H. Song, S. Huang, Y. Dong, and W.-W. Tu, "Robustness and generalizability of deepfake detection: A study with diffusion models," *arXiv preprint arXiv:2309.02218*, 2023.